

"I'm not as afraid as a woman might be about sharing my exact location:" On the Intersection of Identity and Privacy Concerns in Fitness Tracking

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Abstract

Users' perceptions of fitness tracking privacy is a subject of active study, but how do various aspects of social identity inform these perceptions? We conducted an online survey (N=322) that explores the influence of identity on fitness tracking privacy perceptions and practices, considering participants' gender, race, age, and whether or not they identify as LGBTQ*. Participants reported how comfortable they felt sharing fitness data, commented on whether they believed their identity impacted this comfort, and brainstormed several data sharing risks and a possible mitigation for each risk. For each surveyed dimension of social identity, we find one or more reliable effects on participants' level of comfort sharing fitness data, specifically when considering institutional groups like employers, insurers, and advertisers. Further, 64% of participants indicate at least one of their identity characteristics informs their comfort. We also find evidence that the perceived risks of sharing fitness data vary by identity, but do not find evidence of difference in the strategies used to manage these risks. This work highlights a path towards reasoning about the privacy challenges of fitness tracking with respect for the lived experiences of all users.

CCS Concepts

• **Security and privacy** → *Social aspects of security and privacy; Human and societal aspects of security and privacy*; • **Human-centered computing** → **Empirical studies in HCI**; *Human computer interaction (HCI)*; **Empirical studies in ubiquitous and mobile computing**.

Keywords

Fitness trackers, privacy, safety, online survey, identity, risk perception, demographic factors, data sharing, mitigations

ACM Reference Format:

Yeeun Jo, Mahnoor Jameel, Camille Cobb, and Adam Bates. 2025. "I'm not as afraid as a woman might be about sharing my exact location:" On the Intersection of Identity and Privacy Concerns in Fitness Tracking. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 23 pages. <https://doi.org/10.1145/3706598.3713941>

1 Introduction

Fitness trackers allow users to record data about their fitness and exercise activities, monitor their progress, and share results with friends [48]. Made possible through both dedicated wearable devices (e.g., Fitbit [18]) as well as phone-based apps (e.g., Strava [65]), fitness tracking is a booming industry valued at tens of billions of dollars [40, 42]. The widespread adoption of fitness tracking reflects a growing interest in wellbeing and provides a means of empowering people to take control of their physical health.

However, the sensitive nature of health and fitness data means that fitness tracking users are exposed to a host of risks to their privacy, security, and even physical safety. Athletes that track and share outdoor exercise activities may be at risk of being de-anonymized [16], having their home addresses inferred [12, 28], or their exercise equipment stolen [55, 64, 66]. Step count data has been shown to leak information about personality traits [76]. Due to the large amounts of valuable data they amass, fitness tracking vendors have been the target of data breaches [59], and may also be in the business selling user data to third parties [43]. Aggregate fitness data has exposed the locations of secret military facilities [57], and even when anonymized may still be used to violate the privacy of individual users [9].

While users' perceptions of privacy in fitness tracking is already a subject of active study [4, 11, 21, 45], little consideration has been given to the role of social identities in this area. Identity is comprised of the various qualities, beliefs, and perceptual features that characterize a person or group, such as religion, nationality, race, ethnicity, gender, and political alignment, among many others. Members of some identity groups face markedly higher risks through fitness tracking use than others. In the United States, women's access to reproductive health services is being challenged at the federal and state levels, and health tracking data has already been weaponized in some states to restrict access [58]. This reality has



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ACM ISBN 979-8-4007-1394-1/25/04

<https://doi.org/10.1145/3706598.3713941>

already prompted increased scrutiny of the privacy of health and wellness trackers [44, 61]. Members of the LGBTQ* community,¹ especially those of color, are at greater risk of being the victims of physical violence [19]. As fitness tracking exists at the crossroads of technology, health privacy, and real-world safety, it is essential that our approach to fitness tracking is informed by the experiences of those at greater risk.

To address this gap in the literature, we systematically study the role that identity plays in shaping fitness tracking users' risk perceptions, privacy management strategies, and comfort sharing fitness data in different social contexts. We conduct an online survey (N=322) of physically-active United States (US) residents to examine how demographic factors (Gender, Race/Ethnicity, Age, and LGBTQ* community membership) inform participants' comfort with sharing fitness data. We situate the question of sharing fitness data in specific social contexts – with interpersonal (Friends, Family, and Romantic Partner), institutional (Employers, Insurers, Advertisers, and Fitness Tracking Vendors) groups, and the General Public. We also asked participants to brainstorm risks posed by sharing fitness data with these groups, then asked them to identify possible strategies for managing those risks. This survey design engages the following research questions:

RQ1: How does identity inform people's comfort in sharing fitness data across different social contexts?

RQ2: How does identity inform the perceived risks in sharing fitness data across different social contexts?

RQ3: How does identity inform the mitigations suggested to manage these risks?

Our analysis uncovers reliable effects of identity on fitness tracking privacy. To investigate participants' comfort levels in sharing fitness data with different groups, we perform ordinal regression analysis in the Bayesian framework. Notable results include that women are less comfortable than men in sharing fitness data with institutional groups, as are 30-39 year olds (compared to 40+). These findings are also reflected in participants' *perceptions* of how identity affects their comfort level. Qualitative analysis of participant responses reveal that perceptions of vulnerability and privilege strongly influence comfort judgments, with some participants referencing the intersection of multiple facets of identity to explain their answers. When analyzing perceived risks, we find significant differences in the distribution of risks identified by participants of different identity groups (Race and Age). We also find that the risks of fitness tracking are commonly encountered in the real-world, with 58% of participants reporting they had experienced at least one of the five risks they described. However, we do not find evidence that identity informs how privacy risks are mitigated on fitness apps; participants overwhelmingly report that the most effective strategy is to "opt out" of sharing data through various mechanisms (e.g., *Stop Sharing with Party*, *Stop Sharing Data Type*, *Disable/Adjust Privacy Settings*).

This work provides a deeper understanding of the influence of athletes' overlapping social identities on how they engage with the privacy risks of fitness trackers. In discussion, we consider how

our findings reflect the prevailing sociopolitical issues of the day, and consider how methods that are more sensitive than our online survey instrument could be used to disentangle the interleaving effects of identity on fitness tracking. Finally, in light of the disconnect between identity's influence on risk and (apparent) lack of effect on privacy management, we highlight the need incorporate explicitly-inclusive practices into fitness tracking user experience and design research.

2 Background and Related Work

Benefit and Risk of Fitness Tracking Ecosystems. Fitness trackers have been one of the biggest global trends in fitness [68]. A fitness tracking *app*, such as Apple Health or Strava, is a mobile application that can provide health and fitness monitoring dashboards and integrate with social media [2]. Underscoring their popularity, the wearable market size was estimated at USD 39.5 Billion in 2022, while the app market was estimated at USD 6.5 Billion, with both markets expected to grow by approximately 17% annually from 2023 to 2033 [40, 42]. Fitbit and other fitness tracker companies have garnered significant interest from technology companies like Google [50] as well as consumers [34] in recent years. Many apps and wearables collect activity data on a regular basis, such as running routes, heart rate, and step counts and offer premium paid services like body figure analysis. Users can access personalized workout plans tailored to their goals [47], track progress over time [37], and stay motivated through regular exercises [10]. Fitness tracking ecosystems also offer in-depth data analysis, such as calories burned, running routes, sleep quality, or heart rate monitoring, which help users make informed health decisions. Additionally, users can join social challenges to connect with friends and engage in community support [26].

For many, the act of *sharing* fitness tracking data is an important aspect of their exercise routines, creating a sense of shared accountability with other athletes. Despite the enormous benefits fitness trackers offer, their use can involve privacy risks [30, 56, 60]. First, a user may choose to share their data with the tracker's manufacturer and/or the manufacturer's affiliates [71]. Users can also exchange fitness data for benefits provided by third-party organizations (e.g. giving access to insurance providers in exchange for lower premiums [17, 62]). In addition, individuals have the option to post their fitness information on online social networks [4]. Studies have also demonstrated that shared fitness data could be used to deduce users' sensitive information from the data collected by fitness trackers, resulting in disclosure of their online identities [3, 12, 28, 45, 53]. Sharing any kind of health data, such as step counts or heart rate, may leak information about individuals' health status, medical conditions, or even personality traits [76]. While these deductions may be useful to consumers in some situations (e.g., diagnosing COVID-19 [31, 51]), there are a number of privacy and security problems utilizing fitness trackers that have been publicized by the mainstream media [30, 56].

Sharing location-related exercise data, such as the route of a jog or bike ride, is especially fraught with peril. Such data provides a literal map for thieves looking to steal costly exercise equipment (e.g., [55, 64, 66]) and could even jeopardize users' physical

¹CHI Reviewers provided helpful feedback that LGBTQ+ is a more inclusive and widely-accepted term than LGBTQ*. We chose to continue to use the term since it appeared in our survey materials, but wanted to note the limitation.

safety. While fitness tracking services provide a variety of privacy features, including the ability to partially obscure location information, it has been repeatedly demonstrated that users' home addresses can still be inferred by technically skilled [12, 28] or unskilled [45] attackers alike.

The concerns regarding fitness tracking privacy are not exclusively related to sharing with third parties, but also implicate the vendors themselves. In 2016, Fitbit inadvertently disclosed access to customer GPS history as a result of a data breach [59]. Fitness tracking services also sell customer information (e.g., Strava Metro [1]), and users have raised concerns about share-by-default privacy policies in apps. Vendor's sharing practices have even threatened national security: in 2018, it was discovered that Strava's public global heat map of users revealed precise locations of classified military bases, CIA rendition sites, and intelligence agencies [57].

General Digital Privacy Research about Identity. Research on digital privacy and identity has explored factors such as gender, age, and marginalized groups. Gender differences in privacy concerns have been detected, with women often perceiving more risk online and reporting higher levels of privacy concerns than men [5, 25, 49]. Some research has shown that men are more likely to take aggressive behavior and adopt a wider range of behaviors in response to privacy concerns than women [71]. Age has also been found to influence individual perception of information privacy. While older internet users express greater concern (e.g., [5, 25]), younger consumers have lower levels of concern because of the financial benefits that personal information dissemination will bring [14, 25]. Furthermore, marginalized groups, including LGBTQ* groups, face increased privacy risks. These individuals often have heightened concerns regarding the timing and the context of their identity disclosure online [7, 13]. Understanding the specific privacy concern of these demographic identity factors is crucial for developing inclusive and effective privacy measures.

Privacy and Fitness Tracking. Previous research has examined privacy concerns and behaviors of end-users in the context of fitness tracking, exploring aspects such as fitness trackers (e.g., [3, 41, 70]), fitness specific data collection (e.g., [52]), and fitness tracking apps (e.g., [15, 45]). Scholars have investigated users' concerns (e.g., [21, 46, 67]), management strategies (e.g., [4, 74]), and privacy guidance in the fitness tracking domain (e.g., [24, 39, 73]), aiming to counteract privacy risks. Despite the potential risks in engaging with fitness tracking ecosystems, previous studies have demonstrated people believe their gadgets are generally useful [4, 41, 74]. The majority of users, however, are unaware of the privacy issues related to their devices [4, 21, 70, 74], leading to a perception that users are often not very concerned about their privacy [21, 41, 74]. Against this backdrop of privacy and safety concerns, decisions about using fitness trackers and sharing fitness data are highly individualized and situated in relevant social contexts. However, prior work pays limited consideration to social context and even less consideration of how individuals' identity and demographic factors inform their use of fitness trackers.

We set out to address this gap in the literature. We predict differences exist between how fitness tracking privacy is perceived and managed by members of different sociopolitical groups. Such an expectation is supported, in part, by numerous prior studies that

have identified differences in how men and women adopt and use technology [27, 35, 75].

Researchers have also identified a possible relation between demographic identity and privacy. One study provided evidence that men are more aware of publicly-stated fitness tracking apps' privacy policies than women [71]. However, there is a significant gap in the literature concerning the relationship between fitness tracking user's identity and their privacy concerns. There has been little research on the effects of demographic characteristics on fitness tracking users' privacy concerns, and further research is needed to establish whether specific demographic groups of fitness tracking app users have higher information privacy worries than their counterparts.

3 Methods

The aim of this study is to explore how physically active participants' identity informs their perceptions and data sharing practices. To explore the relation between identity and fitness privacy, we conducted an online survey focused on exploring comfort, risks, and mitigation strategies in the context of fitness tracking. We collected data in December 2023, recruiting US-based Prolific users who are highly athletic and who were balanced across the identity facets of Gender, Race, Age, and LGBTQ* status (N=322). Each participant was paid \$5.00 for participating. The survey took a median of 17 minutes and an average of 23 minutes to complete. This compensation was consistent with (and slightly higher than) the U.S. minimum hourly wage and Prolific's suggested rates. The survey contained two attention check questions to determine whether participants were focusing on the survey. Participants were informed of the attention requirements in the consent form, and the questions had clear correct answers. The study protocol was reviewed and approved by the authors' Institutional Review Board (IRB), protocol number #23552 and we obtained informed consent before collecting participants' data.

3.1 Survey Design

We designed the survey to surface the impacts of identity in participants' comfort, risks, and mitigations. The full survey instrument (see Appendix A) asks participants about their comfort with sharing fitness data, potential risks of data sharing, and potential strategies for managing those risks. Throughout the paper, we include the question numbers in Appendix A. The survey flow was as follows:

- We gave participants a definition for fitness trackers with a short story about an athlete named Sam who used a fitness tracking app to record an outdoor run and then posted it on Facebook. We presented the story about a specific fitness tracking scenario to encourage participants to consider both digital privacy (Facebook) and physical safety (outdoor run). We chose the name Sam and used they/them pronouns in the story to be relatively ambiguous regarding the identity facets we were studying (i.e., Gender, Race, Age, and LGBTQ* status).
- We asked participants to imagine that their exercise routines were like Sam's from the short story. We asked them to rate their comfort with "sharing data from your fitness tracker with the following groups of people? [friends, family, romantic partner, employers, insurance company, advertisers, fitness trackers

Gender	Count	is LGBTQ*?	Count
<i>Man</i>	156	<i>Yes</i>	131
<i>Woman</i>	161	<i>No</i>	176
Prefer not to answer	4	Prefer not to answer	10
Prefer to self-describe	1	Maybe	5
Age	Count	Race/Ethnicity	Count
<i>18-29</i>	116	<i>Asian</i>	79
<i>30-39</i>	107	<i>Black</i>	117
<i>40+</i>	99	<i>White</i>	123
		Native Hawaiian or Pacific Islander	1
		American Indian or Alaska Native	1
		Prefer not to answer	5
		Not listed above	5

Table 1: Summary of participant self-reported demographic information ($N=322$)—We used Prolific’s screening criteria to collect a best-effort balanced sample for the italicized identity characteristics. However, to ensure accuracy and respect participant privacy we then used participants’ self-reported demographics information in our analysis.

vendors, and general public]” on a 5-point Likert scale ranging from extremely uncomfortable (1) to extremely comfortable (5) (Q2.(a) - (h)). We refer to the 8 groups with whom one might share data as *Sharing Parties*, and we refer to this set of questions as the *Group Comfort Questions*.

- We asked if participants thought their identity (i.e., Gender, Race or Ethnicity, Age, and whether or not they are LGBTQ*) informed their responses to [all, some, or none] of the *Group Comfort Questions*. They could also indicate “I’m not sure” (Q3, 5, 7, 9). We also asked them to explain the rationale behind their answers (Q4, 6, 8, 10). We refer to these items as *Identity Comfort Questions*.
- We asked participants to complete a brainstorming activity in which they generated 5 potential risks of sharing data from fitness trackers. For each risk, participants selected one of the 8 *Sharing Parties* from the prior questions, described the risk (free response), and gave a rating of its likelihood (extremely unlikely (1) to extremely likely (5)) and harmfulness (not at all harmful (1) to extremely harmful (5)) (Q11.(a)-(d)). We refer to these items as *Risk Brainstorming Questions*.
- We re-displayed each risk the participant had brainstormed one at a time. For each risk, we asked whether the risk had ever actually happened to the participant (Q12) and what strategies they could use to manage the risk (Q13). We refer to these items as *Mitigations Questions*.

3.2 Recruitment

We used the online crowdsourcing platform Prolific to recruit participants. We advertised the study as “a research study to understand users’ experiences and concerns around fitness-tracking apps (e.g., Google Fit, Strava) and fitness trackers (e.g., Fitbit, Apple Watch).” We made heavy use of prescreening criteria to achieve an appropriate participant sample. The survey was available only to US-based Prolific users who exercise at least 150 minutes per week. For statistical power and to get a range of perspectives, we wanted a sample with roughly the same number of participants in each demographic

group. We used Prolific’s “Ethnicity - simplified” criteria for Race and restricted to Asian, Black, and white individuals. We bracketed age into three groups (18-29, 30-39, 40+). For gender, we used Prolific’s “Gender” screening criteria, which explicitly includes trans man/woman with their correct gender and has an option of non-binary. Our screening criteria did not include non-binary people. We used Prolific’s “LGBTQ*” screening criteria, which presents a binary choice. We manually balanced across the different identity facets by releasing the survey in 36 identity-screened batches. For example, one batch screened for US-based participants who exercise over 150 minutes per week, are a woman, are Black, are between 18-29 years old, and are not LGBTQ*. Each of the 36 batches (2 Genders * 3 Races * 3 Age buckets * 2 LGBTQ* status options) targeted 10 participants for a total target of 360 participants. Despite leaving batches open for several weeks, several batches did not fill (i.e., [Asian LGBTQ* Men (in all age groups)] and [Asian LGBTQ* Women in age groups 30- 39 and 40+]) due to low numbers of active users meeting criteria.

We also collected self-reported demographic data from participants, which we choose to use in our analysis instead of Prolific’s prescreening demographic data. The demographics of the resulting sample are summarized in Table 1. If a participant indicated that they preferred not to answer a demographic question, or if their response did not fit into the group comparisons specified in our analysis plan, their data was excluded from that portion of the analysis. For example, 15 participants preferred not to indicate an LGBTQ* status, so we do not include their responses in comparisons between the LGBTQ* and Non-LGBTQ* groups.

These choices explained above were informed by preliminary data collection on Prolific where we recruited a US Representative sample of participants that exercised at least 150 minutes per week. In the preliminary survey, over 88% of participants reported using a fitness tracker, indicating widespread adoption in this sample population. The demographic group with the lowest adoption rate, Black participants, was still high at 82%. Thus, we determined that the exercise requirement was an excellent proxy for exposure to fitness trackers. By oversight, we omitted the question about participants’ use of fitness trackers in our final survey (See 3.4).

The preliminary survey also informed our choices to collect a manually balanced sample that would support our data analysis plan. A demographic summary for preliminary data collection can be found in Appendix C). Manually balancing the sample addressed the issue of imbalance between LGBTQ* (64 Yes, 339 No) and Race/Ethnicity (44 Asian, 33 Black, 310 white) group representation. Unfortunately, the preliminary responses and subsequent experimentation also revealed that Prolific’s user pool was not adequate to collect data for some race and ethnicity groups, such as Pacific Islanders and Latinx individuals, when our prescreening exercise criteria was applied. As a result, we chose to only recruit participants identifying as Asian, Black, and white in our primary data collection.

3.3 Data Analysis

3.3.1 Data Quality Checks. We examined all responses to determine if any data should be excluded from analysis. Although we included two attention check questions, no respondents failed

both, and only three failed the second. We flagged these responses as needing additional scrutiny and turned to other data quality checks. Since we recruited participants in batches using Prolific’s screening criteria and also collected demographic information in our survey, we had two sources of demographic data. We checked that both sources matched, and we found discrepancies in 49 cases, mostly minor. Minor discrepancies included, for example, respondents electing not to report some demographic information in our survey, making selections in our survey’s race/ethnicity choices in a way that did not map cleanly back to Prolific’s simplified race categories, misaligned LGBTQ* status, off-by-one errors between Prolific’s age buckets and the numerical self-reported age, and a couple of participants whose self-reported and Prolific gender records conflicted but who also reported being LGBTQ*.

Finally, we examined all responses holistically, paying special attention to the quality of free-response answers and giving specific scrutiny to the responses flagged as described above. We agreed that all three respondents who had failed a single attention check had other indications that their responses were in good faith and so did not exclude any of these responses. The respondent with more major demographic discrepancies had also given low-quality free-response answers; we excluded their response. None of the other responses with minor demographic discrepancies had signs of being low-quality, so we included their data and used self-reported demographic data in all of our analyses.

In our holistic examination of survey responses, we identified six other responses that were such low-quality that we agreed they should be excluded (i.e., the first, third, and fourth authors came to this consensus about this decision through synchronous discussion). We explicitly aimed to be conservative about dropping data due to this manual examination; for example, although there were some free responses for which respondents had clearly misunderstood our question, we included these response as long as we believed that they could have been given in good faith (and dealt with these misunderstandings in our later qualitative analysis procedures). We retained 322 valid responses for our study.

3.3.2 Quantitative Analysis. To test for the presence of identity differences on sharing comfort levels, we perform proportional-odds logistic regression modeling using a Bayesian framework. Bayesian analysis is gaining popularity in the HCI community due to its ability to report more precise effects and draw more principled conclusions from small studies [36]. In contrast to frequentist null hypothesis significance testing, which maps the research into a binary question (reject or fail to reject the null hypothesis), Bayesian analysis instead directly estimates the effect size in the form of a mean and 95% *credible interval*. Further, because there are no null hypotheses or alpha values, it is not necessary to adjust for multiple comparisons when performing Bayesian analysis. For these reasons, Bayesian analysis offers many advantages even when there is no basis for specifying an informative prior, as is the case in our work.²

For each of the 8 measurements from the Group Comfort Questions, corresponding to the 8 Sharing Parties (Q2. (a)-(h)), we perform a proportional-odds logistic regression. We then specify the following identity factors as predictor variables, using weighted

contrasts to control for uneven numbers of observations: Gender (nominal; Levels: Men, Women), LGBTQ* status (nominal; Levels: False, True), Race=Asian (nominal; Levels: False, True), Race=Black (nominal; Levels: False, True), Race=White (nominal; Levels: False, True), Age Group (ordinal; Levels: 18-29, 30-39, 40+). Comparisons within Age Group are nested (18-29 vs. 30+, 30-39 vs. 40+).

To estimate our models, we use the Stan software package and its R front-end, brms.³ Bayesian estimation is performed with the brm function with the cumulative (“probit”) family using the Markov Chain Monte Carlo (MCMC) sampling algorithm. Each model is fit using four Markov chains with 2,000 iterations per chain, the first 1,000 of which are “burn in” samples that are discarded. Model fit diagnostics suggest that all the chains converged in our experiments ($\hat{R}=1$ for all estimates). We use the default priors for regression modeling, which are uninformative. This reflects, for example, the prior belief that men and women are equally comfortable sharing their fitness data with friends.

Though our statistical analyses consider each Sharing Parties separately, figures and text in Section IV combine these parties into Interpersonal Groups (i.e., Friends, Family, Romantic Partner) and Institutional Groups (i.e., Employers, Insurance Company, Advertisers, Fitness Trackers Vendors). We also use the shorthand of Vendors and Public.

In analyzing the Risk Brainstorm Questions, we employed the chi-square test (χ^2) to investigate whether there exist statistically significant variances in responses within grouped identity. We specify the predictor variables the same as we did for analyzing the Group Comfort Questions. To account for the potential of type I errors arising from multiple comparisons, we apply the False Discovery Rate-Benjamini and Hochberg (FDR-BH) [6] to compare counts of the risk types that participants had brainstormed at least once. Post-hoc comparisons were corrected for multiple-testing using the Benjamini and Hochberg method.

We also analyze the full distribution of social contexts within each type of risk, beyond simple frequency counts or mode categories to include insights from less common occurrences across the entire range of responses.

3.3.3 Qualitative Analysis. We separately analyzed responses to (a) the question, “How does your identity (i.e., Age, Gender, Race or Ethnicity, LGBTQ* Status) impact your comfort level?” (Q4, 6, 8, 10) and (b) the risks and corresponding mitigations participants wrote in (Q11, 13). For (a), our unit of analysis was one participant’s response to why or how one facet of their identity informs their comfort, and we additionally referred to the participants’ responses to Group Comfort Questions (Q2) and Identity Comfort Questions (Q3, 5, 7, 9), and use their demographic data (per the batch screening criteria from Prolific) to contextualize their response. For example, when P6 wrote “Same as before..” in explanation of whether her age informed her comfort, we looked back at her previous response; when P320 wrote “I’m not sure I can feel more uncomfortable about the idea of people being able to see my running,” noticing that all of his Group Comfort Questions responses were scores of (1) or (2) (i.e., extremely uncomfortable or slightly uncomfortable) helped us be confident in our understanding of what he wrote. For (b), we again incorporated Group

²When an informative prior is not set, the mean effect in a Bayesian regression will be nearly identical to the estimate in a Frequentist regression.

³<https://mc-stan.org/users/interfaces/brms>

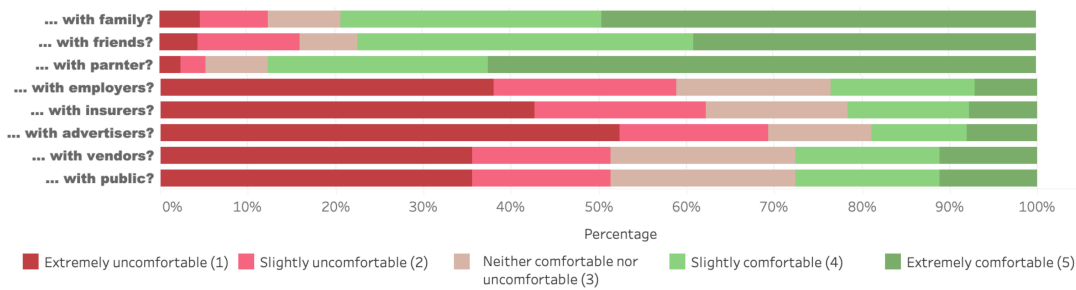
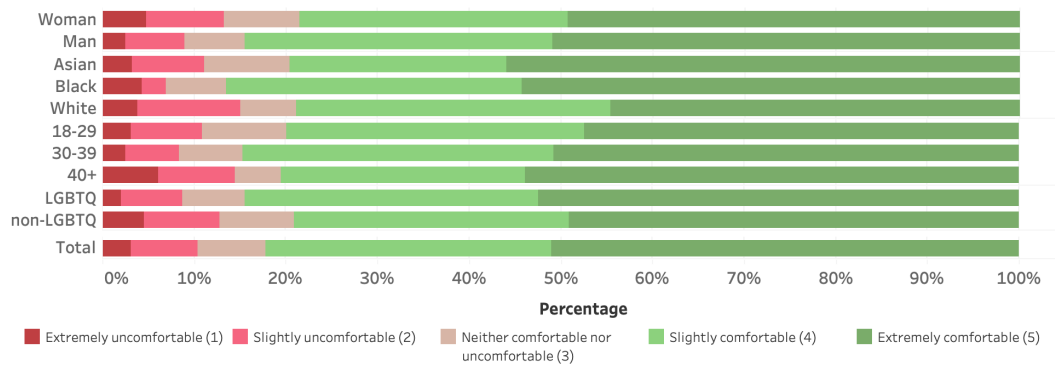
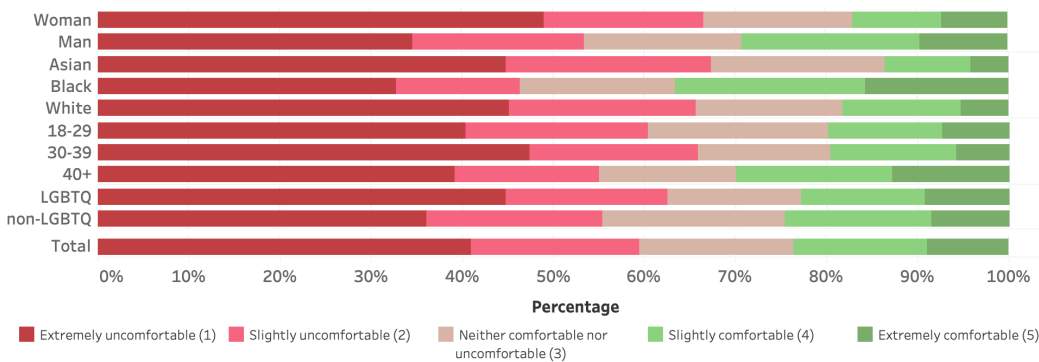


Figure 1: How comfortable would you be sharing data from your fitness tracker...? (Q2)– Overall, participants were much more comfortable sharing fitness data with Interpersonal Groups than Institutional Groups.



(a) Interpersonal Groups (Friends, Family, Romantic Partner)



(b) Institutional Groups (Employers, Insurers, Advertisers, Vendors)

Figure 2: Differences in sharing comfort by identity (Q2)–Responses collapsed across the Interpersonal Groups and Institutional Groups. Women appear to be less comfortable sharing fitness data with Institutional Groups.

Comfort Questions (Q2). and demographic data, as well as the Identity Comfort Questions (Q3, 5, 7, 9).

The research team started by familiarizing ourselves with the data by reading through responses, which overlapped with our data quality checks, as described above 3.3. We first open coded the responses; the first author open coded all responses and third author (for (a), and second author for (b)) open coded a subset of responses (approximately 5-10%). The research team met and collaboratively discussed the open codes, grouping them into higher-level themes and writing corresponding definitions. This conversation resulted in the incorporation of these higher-level topics and meanings into

codebooks ready to apply on (a) and (b). Because we anticipated that some individual free response answers were still unclear or incoherent (despite the decision in data quality checks to not exclude the survey response), we designed our codebooks to code for instances when the response clearly and unambiguously met the coding criteria. Thus, low-quality free response answers simply do not have codes applied to them.

Each codebook was independently applied by two authors (i.e., first and third author for (a), first and second author for (b)) to all of the data. We calculated Inter-rater reliability (IRR) and it ranged depending on almost perfect to fair/modestly perfect. Since not all

Are you comfortable sharing your data with...									
Variable	Friends?			Family?			Partner?		
	Mean	sd	CI	Mean	sd	CI	Mean	sd	CI
<i>Gender: Woman</i>	-0.19	0.13	[-0.44, 0.06]	-0.19	0.13	[-0.45, 0.07]	0.04	0.14	[-0.23, 0.32]
<i>Asian</i>	0.21	0.31	[-0.42, 0.81]	0.03	0.32	[-0.60, 0.63]	0.50	0.32	[-0.14, 1.11]
<i>Black</i>	0.40	0.30	[-0.18, 1.00]	0.23	0.31	[-0.38, 0.81]	0.43	0.31	[-0.18, 1.05]
<i>18-29</i>	-0.03	0.14	[-0.31, 0.25]	-0.20	0.14	[-0.48, 0.07]	-0.17	0.15	[-0.45, 0.13]
<i>30-39</i>	-0.02	0.16	[-0.34, 0.29]	-0.19	0.17	[-0.52, 0.13]	0.07	0.18	[-0.26, 0.42]
<i>LGBTQ*</i>	-0.10	0.13	[-0.34, 0.15]	-0.23	0.13	[-0.49, 0.02]	-0.07	0.14	[-0.35, 0.21]

Table 2: Ordinal regression models in the Bayesian Framework. (Interpersonal Groups)—The unit for estimate (Mean) and standard deviation (sd) is 5-point Likert scale. Bayesian analysis estimates effect size as the 95% Credible Interval (CI) of the posterior distribution. Shaded cells indicate a reliable effect where the entire credible interval suggests an increase (green) or decrease (red) in comfort. For example, we observe that Black participants are consistently more comfortable when sharing with institutional groups than other participants.

Are you comfortable sharing your data with...												
Variable	Employers?			Insurance Company?			Advertisers?			Vendors?		
	Mean	sd	CI	Mean	sd	CI	Mean	sd	CI	Mean	sd	CI
<i>Gender: Woman</i>	-0.41	0.13	[-0.66, -0.16]	-0.28	0.12	[-0.52, -0.03]	-0.31	0.13	[-0.56, -0.06]	-0.39	0.13	[-0.65, -0.14]
<i>Asian</i>	0.59	0.33	[-0.06, 1.25]	0.70	0.37	[-0.01, 1.44]	0.80	0.42	[-0.00, 1.65]	0.68	0.35	[0.01, 1.37]
<i>Black</i>	0.99	0.32	[0.35, 1.64]	1.16	0.36	[0.46, 1.88]	1.40	0.41	[0.62, 2.25]	1.45	0.35	[0.78, 2.15]
<i>18-29</i>	0.02	0.13	[-0.24, 0.28]	0.09	0.14	[-0.19, 0.36]	0.08	0.14	[-0.20, 0.37]	-0.02	0.14	[-0.29, 0.24]
<i>30-39</i>	-0.16	0.16	[-0.47, 0.15]	-0.40	0.16	[-0.72, -0.09]	-0.40	0.17	[-0.72, -0.08]	-0.20	0.15	[-0.51, 0.10]
<i>LGBTQ*</i>	0.09	0.13	[-0.16, 0.34]	-0.02	0.13	[-0.26, 0.23]	0.29	0.13	[0.03, 0.55]	0.09	0.13	[-0.16, 0.33]

Table 3: Ordinal regression models in the Bayesian Framework. (Institutional Groups and Public)—The unit for estimate (Mean) and standard deviation (sd) is 5-point Likert scale. Bayesian analysis estimates effect size as the 95% Credible Interval (CI) of the posterior distribution. Shaded cells indicate a reliable effect where the entire credible interval suggests an increase (green) or decrease (red) in comfort. For example, we observe that Black participants are consistently more comfortable when sharing with institutional groups than other participants.

IRR scores were as high as expected, the two coders discussed all disagreements and came to a consensus about most. When consensus was still not reached between the two coders, the response was discussed with the last author. In this way, we reached consensus about all responses and codes, and this finalized dataset is the data underlying Section 4.

3.4 Limitations

Our study design has several inherent limitations, which readers should consider as they interpret our findings. First, the prescreening criteria and number of active users on Prolific were limiting. Even in the data we did collect, some batches did not meet their target participant count. Correspondingly, since our planned statistical analyses relied on distinct separations between the identity facets of Gender, Race, Age, and LGBTQ* status and of the levels within each of these facets, there are groups of people that we did not reach. For example, we did not include participants who recorded their ethnicity as “Mixed” on Prolific, or any of the ethnicity categories from Prolific’s non-simplified version of that demographic screening criteria. This includes the Hispanic/Latino ethnicity, which is the second largest ethnic group in the United States, because this was not in Prolific’s simplified ethnicity criteria. Additionally there are Hispanic/Latino people who are also white and/or Black, which, again, our statistical analyses could not handle.

We acknowledge that this type of oversimplification is one of the key limitations of any quantitative analysis about people and

their identities; thus, the qualitative perspectives surfaced in this work are vital for getting a fuller picture. We also emphasize that research that takes a fully qualitative approach is equally important. We believe that despite these limitations, our quantitative analysis offers useful and novel insights in this space. There are also other identity facets and personal characteristics that we did not measure or balance for in our sample and for which Prolific’s user base may not be representative of the general population. For example, our sample is more likely to be educated and skilled in technology by virtue of recruiting from an online crowdsourcing platform. We cannot know the extent to which our sample of Prolific users is representative of the overall population of fitness tracking users.

4 Data and Findings

4.1 How does identity inform people’s comfort in sharing fitness data across different social contexts?

In keeping with findings from prior work [21], the comfort levels that participants reported depended on who their fitness tracking data would be shared with. Shown in Figure 1, *participants tend to be much more comfortable sharing with Interpersonal Groups (Friends, Family, Romantic Partners) than Institutional Groups (Employers, Insurers, Advertisers, Vendors)*. Figure 2 shows data aggregated into Interpersonal Groups vs. Institutional Groups. While demographics are an oversimplified approximation of identity, we observe

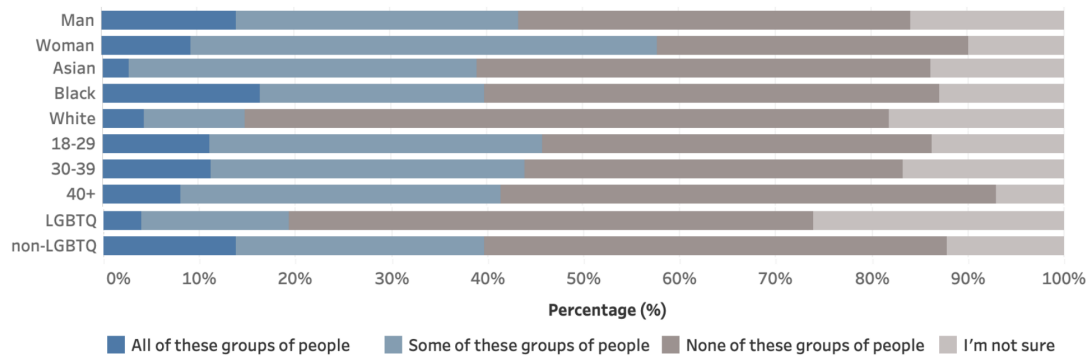


Figure 3: Do you think your Identity (i.e., Gender, Race or Ethnicity, Age, LGBTQ* Status) informs your comfort in sharing data with any of the above groups of people? (Q3, Q5, Q7, Q9).

My identity makes me...	More comfortable	Less comfortable	No direction indicated
More at risk	11 responses, 7 participants	105 responses, 72 participants	31 responses, 29 participants
Less at risk	36 responses, 29 participants	1 response, 1 participant	43 responses, 36 participants
No direction indicated	28 responses, 24 participants	53 responses, 45 participants	-

Table 4: Comparing Comfort and Risks in Relation to Identity—Participants frequently explained how their identity informed their comfort (or likelihood of sharing) and/or risk in their free-response answers Q4, Q6, Q8, and Q10. 153 (47%) of participants indicated a comfort or risk direction for at least one response. Very few participants/responses indicated that higher risk made them more comfortable sharing, or vice versa.

demographic variations in comfort levels, which suggests that identity may impact comfort.

To confirm this observation, we conducted a Bayesian ordinal regression for each identity group using an uninformative prior, resulting in 8 models (one per party) that use identity facets as predictor variables. Models for the 3 interpersonal groups are found in Table 2, with models for institutional groups in Table 3. While we observe no major effects of identity on interpersonal sharing, numerous effects are found within institutional groups. Most notably women and 30-39 year olds are found to be less comfortable sharing with most/all institutional groups when compared to men and 40+ year olds, respectively. In turn, Black participants were consistently more comfortable sharing with institutional groups as compared to non-Black participants.

Participants also self-reported whether different identity characteristics informed their sharing comfort levels (Q3, Q5, Q7, Q9), summarized in Figure 3. Unlike the above analysis, participants' self-reported impact in these questions was non-directional and was asked once per identity characteristic (not per-party). 64% of participants reported at least one of their identity characteristics informed their comfort. Gender was the characteristic for which the

highest proportion of participants indicated it informed their comfort for some or all groups (50% for gender, 45% for age, 30% for race/ethnicity, and 25% for LGBTQ* status).

To understand more about how identity matters, we surface results from participants' follow-up explanations of how or why their identity characteristics did or did not inform their comfort (Q4, Q6, Q8, Q10).

4.1.1 Direction of Impact. Effects observed in our regression analysis were confirmed in many participants' free response explanations, which we coded based on whether participants said they felt they were more or less at risk and whether participants said that they were more or less comfortable sharing (or more or less likely to share) fitness tracking data. The results of this coding are shown in Table 4. For example, P312 described the impact of his race as follows: "As a black person I am less comfortable sharing my location data due to fear of being attacked." Of the 118 participants who reported that their identity made them less comfortable (or likely) to share (the data from fitness trackers), 72 participants elaborated they felt less safe and more at risk. Fewer participants reported feeling more comfortable with sharing data (60 participants, 19%; 76 responses). For example, P255 wrote, "I am white, so for similar reasons as above as to why I would feel more comfortable in sharing

*my physical running location that someone who is black, for example. We all know what happened to Ahmaud Arbery.”*⁴

In contrast with P255’s quote above, 7 of the 60 participants, who said that their identity made them more comfortable or likely to share fitness tracking data, explained that they felt less safe or more at risk. For example, P191 wrote, *“As a female, I definitely feel safer if I am able to share my data with my family, friends, and romantic partner just in case anything should happen to me.”* That is, we found that feeling (or being) less safe could have divergent impacts on sharing comfort. For example, P192 wrote, *“there are some groups I don’t want to have this information because I think it might compromise my safety (general public, employers), but then other trusted people that I do want to share with because it makes me feel more safe.”*

On the other hand, 97 participants (30%; 151 responses) wrote free-response answers that conveyed their belief that a particular identity characteristic did not or should not matter in terms of its impact on comfort, or that they could not understand why it would for anyone, not just themselves. For example, P103, a white man who is over 40 years old and not LGBTQ* wrote, *“I don’t think it really matters”* about the impact of gender, *“race has no bearing on anything in society,”* *“why would this matter?”* about the impact of being LGBTQ*, and *“age is also irrelevant.”*

4.1.2 Privilege Appears to Inform Comfort. Especially among self-reported effects of identity characteristics on sharing comfort (Figure 3), we can see that responses are consistent with sociopolitical privilege, with more privileged identity groups (Men, White People) reporting less of an impact less privileged or more vulnerable groups (Women, People of Color). In other words, *Participants were more likely to indicate that an identity characteristic informed their comfort if that identity occupies a lower tier of the privilege hierarchy.* The fact that identities enjoying greater sociopolitical privilege were less likely to cite identity as an influence on their comfort may also be interpreted as further evidence that privileged identity groups are often ascribed as a “default” identity. For example, only 14% of white participants thought that their race/ethnicity informed their comfort levels, as compared to 38% of Black participants. 7 participants explicitly linked the concepts of vulnerability and/or privilege to their comfort with or likelihood to share fitness tracking data.

Of course, not all participants associated with less-privileged identity groups responded consistently with this trend, with some even pushing back on the relevance of such narratives to fitness tracking. For example, P136, a Black person, explaining that his race did not inform his comfort, wrote, *“I am African American and choose not to acknowledge and perpetuate the progressive left’s damaging and unhelpful language, which includes the concept of ‘race.’ ...There is one race, the human race.”* P254, a queer person, responded to the question about whether his LGBTQ* status informed his comfort with, *“who I am attracted to has nothing to do with my concern for privacy.”*

Even though we asked separately about the impact of participants’ gender, race/ethnicity, age, LGBTQ* status, 27 participants wrote explanations that re-emphasized other aspects of their identity in addition to the one we had asked about. Most often, they talked about the combined impact of their gender and age or gender and race. For example, P312 wrote *“As a black person I am less comfortable sharing my location data due to fear of being attacked”* in response to our question how his race/ethnicity informed his comfort with sharing and then noted that his gender has the inverse impact on his comfort: *“Taking race out of the question, being perceived as male means that I’m much less likely to be targeted by stalkers”* We consider this response to be indicative of the intersectionality of experiences related to different facets of identity.

136 participants (42%; 270 responses) described explicit risks when explaining their answers, including “stalking” (P308), “identity discrimination” (P319), “being targeted for a hate crime” (P96), “body shaming” (P210), “being kidnapped” (P41), “safety” (P206), and “information used for malicious reasons due to identity” (P176). This unprompted association between high-level comfort with specific risk concerns validates our decision to have participants brainstorm potential risks of sharing data later in the survey, which we analyze in Section 4.2.

4.1.3 Other Factors Also Inform Comfort. 209 participants (65%; 427 responses) mentioned other factors besides risk or vulnerability that they saw as relevant to their response. For example, many participants felt that their comfort was primarily affected by their personality or inherent disposition toward valuing privacy: *“I am a very private person and regardless of gender I am not comfortable sharing this info with most people”* (P218) vs. *“I don’t think my gender does [inform my comfort], I think my laidback attitude towards data privacy does”* (P32). P232 likewise emphasized his relationship to privacy, but conveyed a belief that there are gender-correlated differences in this: *“suppose that women may value privacy more, but I’m a man and I think men can value privacy a lot too.”*

Athletic ability or performance was another factor that many participants emphasized as important to their comfort with sharing. For example, many were more comfortable sharing if they felt good about their performance (or vice versa), such as P6 who wrote: *“I don’t feel like I experience anything specific to being female.. I think it would be embarrassment if my stats shared weren’t good enough.”* However, other participants noted that the social pressures that drive this mindset may also be connected to identity. For example, P180 wrote *“I don’t want my mother to make sabotaging comments about my weight related to the tracker. She isn’t invested in my male siblings weight in the same way. That’s related to gender.”*

Other factors that participants mentioned included: characteristics of the group that data is being shared with (e.g., your closeness or similarity with them, their trustworthiness); which types of fitness data specifically they are sharing (e.g., exact location in real time vs. other types of data); comfort or competence with technology, which many saw as correlating with age; exercise habits that affect risk (e.g., exercising alone, habitually running the same routes); societal issues that affect vulnerability (e.g., being insecure about insurance or employment, living or exercising in locations that are less safe for some people – *“if I lived in an area where I*

⁴In 2022, Arbery — a Black man — was murdered while he was jogging through a predominantly white neighborhood [54]. There is not a connection in this case to fitness tracker data specifically increasing risk; rather, it is a reference to the broader riskiness of “Running while Black” [72].

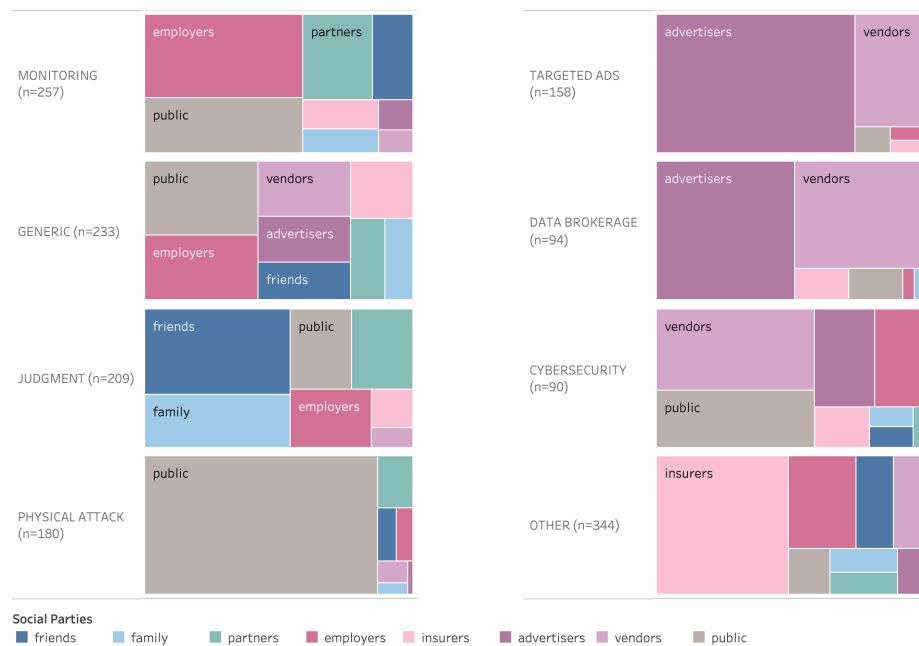


Figure 4: When sharing data from fitness trackers with [Sharing Parties] a risk that might arise is [Risk] (Q11)—Note that n indicates the number of risk responses written in the Brainstorming Questions (Q11.(b)) For example, participants wrote in 287 Monitoring risks.

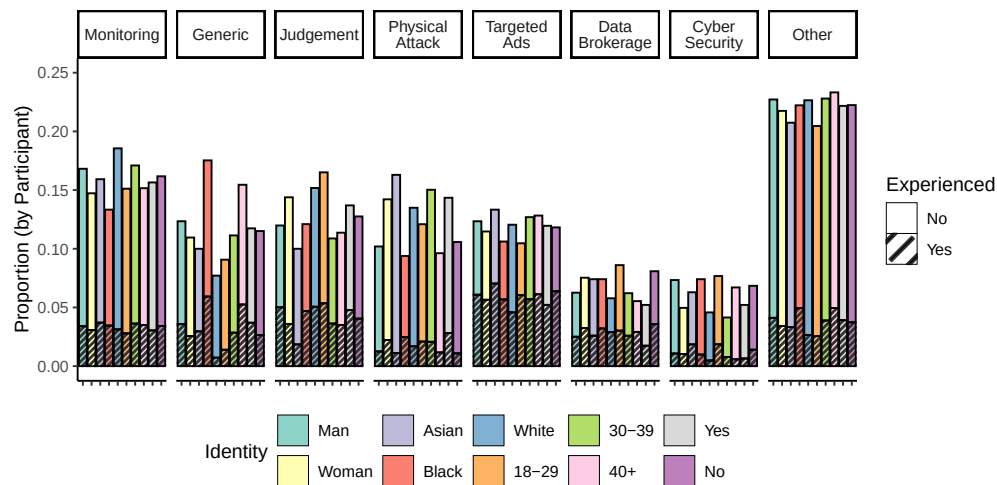


Figure 5: Differences in risks by demographics (Q11)—with risks compared across different identity groups. Identity Factors are denoted with bar patterns while Identity Levels are color-coded. To simplify comparison between factors with 2 levels (Gender, LGBTQ) and 3 levels (Race, Age), we report the proportion of participants for a given Identity Level that at least once identify a given risk. We confirmed that the pattern was similar when compared with the differences in risks by demographics in the proportion of risk items.

felt somewhat persecuted for my race, I would probably feel less comfortable” (P164)); past experiences (e.g., “I was followed by a strange man once while jogging and I don’t want to open myself up to that kind of experience” (P205)); perceptions about how aspects of their

identity beyond those we asked about directly affect their vulnerability, risk, or comfort (e.g., P117 emphasized being a “random nerdy guy,” suggesting that more high-profile people like celebrities or politicians may be at greater risk than he is), and 11 LGBTQ participants expressed that being “straight passing” (phrasing used*

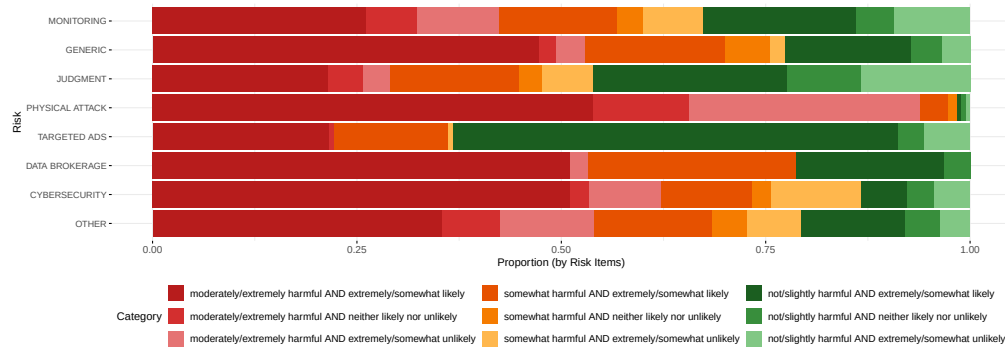


Figure 6: This risk is [Select Likelihood]. If it did happen, it would be [Select Harmfulness] (Q11(c)-(d)).

by P320 and P322) minimized any increased vulnerability due to this facet of their identity.

4.2 How does identity inform the perceived risks in sharing fitness data across different social contexts?

Each participant brainstormed five potential risks that could emerge from sharing fitness tracker data with the various sharing parties, identifying the party the risk pertained to (Q11.(a)), the risk itself (Q11.(b)), the risk’s likelihood (Q11.(c)), the harmfulness of the risk if it were to occur (Q11.(d)), and whether they had actually experienced this risk (Q12). We used our codebook to categorize a total of 1610 risk descriptions into 7 types of risks: Monitoring, Generic, Judgment, Physical Attack, Targeted Ads, Data Brokerage, Cybersecurity, along with a category for Other less frequent appearing risks. Appendix B describes each risk in more detail.

4.2.1 Participants Perceive Numerous Risks. Figure 4 illustrates the distribution of risk categories and the associated sharing parties. All 7 of the identified themes appear regularly among participants’ responses, with frequencies ranging from 22% to 50% per participant per theme. While Generic risks describing a vague privacy concern were common (117 participants, 36%; 233 responses), the most frequent risk identified by participants related to Monitoring (162 participants, 50%; 257 responses), which expresses concerns about being observed or having one’s whereabouts and personal activities under scrutiny.

Figure 4 also provides a breakdown of each risk by social party. In keeping with findings from prior work [74], *participants were generally sensitive to social contexts, perceiving risks as varying depending on the specific social environments in which they were situated.* For example, we can see that the next most frequent risk, Judgment, was predominantly associated with Interpersonal Groups like Friends and Family, whereas Physical Attack, Targeted Ads, Data Brokerage, and Cybersecurity risks were more frequently linked to Institutional Groups (e.g., Advertisers, Vendors). On the other hand, Monitoring and Generic risks were distributed across all eight sharing parties. Risks that fell into the Other grouping were either too infrequent among responses or were exclusively

Identity Comparison	χ^2	df	p-value	adjusted p-value
Gender: Woman vs. Man	7.17	7	0.4116	0.5243
Race: Asian vs. Others	6.81	7	0.4494	0.5243
Race: Black vs. Others	24.51	7	0.0002	0.0065*
Race: White vs. Others	17.85	7	0.0127	0.0443*
Age: 18-29 vs. 30+	16.2	7	0.0234	0.0545*
Age: 30-39 vs 40+	6.93	7	0.4367	0.5243
LGBTQ: no vs. yes	4.39	7	0.7344	0.7344

Table 5: χ^2 tests results on the proportion of brainstormed risks (by participant) (Q12)—We test for differences in the proportion of brainstormed risks by identity characteristic using χ^2 tests of independence. The * in the rightmost column indicates significant differences between comparison groups ($p < 0.05$). Adjusted p-values represent the results of applying FDR-BH to address multiple comparisons.

associated with a single social context. Most notably, loss or increased costs of insurance account for 60% of the risks categorized as Other and were attributed to insurers.

4.2.2 Perceived Risks Vary by Identity. When comparing identified risks across different demographic factors (Figure 5), we find evidence that *identity impacts risk perception*. The most pronounced identity-related difference was observed among Black participants, who reported Generic risks nearly twice as often as their white or Asian counterparts. Specifically, 15.8% of Black participants’ risks were categorized as Generic, compared to 8.9% for Asian and 7.6% for white participants’ risks. Across nearly all risk categories, the most substantial differences in proportions were associated with race and age dimensions. For instance, Generic, Monitoring, and Physical Attack varied notably by race while Physical Attack, Judgment, Data Brokerage, Targeted Ads, and Cybersecurity risks exhibited significant variation across different age groups. This highlights how different aspects of identity, including race and age, influence the perception and reporting of various types of risks.

To confirm these observations, we conduct χ^2 tests that compare the proportion of brainstormed risks of each category between different identity characteristics. Specifically, using the results from Figure 5 we create a vector for each identity group that includes the 7 proportions by participant observed for each risk type (excl. Other), then conduct the same group comparisons from our regression model. To account for multiple post-hoc comparisons, we report adjusted p-values using the Benjamini-Hochberg method. Results are summarized in Table 5. We confirm that the distribution of brainstormed risk responses is significantly different between Black participants and other participants, white participants and other participants, and 18-29 and older age groups.

4.2.3 Participants Believe Risks are Likely and Harmful. In general, almost all participants input at least one risk that they felt was extremely or somewhat likely (311 participants, 96%; 1100 responses). 295 participants also wrote in at least one risk that they felt would be extremely or somewhat harmful if it were to occur (91%; 792 responses). Examining aggregated results across all participants, we observe variation in both the likelihood and harmfulness of the different risk types. Figure 6 demonstrates this in a composite of likelihood and harmfulness of each risk. For example, Physical Attack is nearly unanimously scored as moderately/extremely harmful 94%, while just 22% view Targeted Ads as equally harmful. Although Monitoring was mentioned by the highest number of participants, it was rated unlikely and/or not harmful by some, with 27% of responses indicating it as extremely or somewhat unlikely and 33% as slightly or not at all harmful. Targeted Ads was often perceived as especially not harmful, with 63% of responses categorizing it as slightly or not at all harmful. On the other hand, Data Brokerage had the highest proportion of likelihood ratings, with 78 participants (24%; 92 responses) rating it as extremely or somewhat likely.

We examine identity-based differences in harmfulness in Appendix C. We refrain from conducting further null hypothesis significance tests due to the sheer number of possible comparisons, and instead remark only upon the raw distributions of responses. *We observe demographic variations in harmfulness and likelihood levels, which suggests that identity does impact the perception on risks.* For example, more 40+ age groups perceived the risks more harmful and likely compared to others, as are the Black participants compared to others and women were more likely to report very harmful types of risks compared to men.

4.2.4 Experienced Harms from Risks. We also asked participants if they had experienced the risks they had brainstormed (Q12). More than half of participants (187, 58%; 370 responses) indicated that they had experienced at least one of their brainstormed risks. As shown in Figure 5, at least one participant had previously encountered all of the risks presented. For example, P85, a Black woman, reported experiencing all five of the risks she identified, including Generic, Monitoring, and Targeted Ads risks. 27 participants (31 responses) evaluated the risks are unlikely but indicated they have experienced such risks before. For example, P27 stated a risk related to romantic partner is “jealousy or competition” and evaluated it as somewhat unlikely but indicated they have experienced it before.

Identity Comparison	χ^2	df	p-value	adjusted p-value
Gender: Woman vs. Man	3.2	5	0.6694	0.9624
Race: Asian vs. Others	2.85	5	0.7223	0.9624
Race: Black vs. Others	9.26	5	0.0992	0.6943
Race: White vs. Others	3.47	5	0.6284	0.9624
Age: 18-29 vs. 30+	1.82	5	0.8741	0.9624
Age: 30-39 vs 40+	1	5	0.9624	0.9624
LGBTQ: no vs. yes	2.15	5	0.8181	0.9624

Table 6: χ^2 tests results on the proportion of brainstormed mitigations (by participant) (Q13)—We test for differences in the proportion of brainstormed mitigations based on identity characteristic using χ^2 tests of independence. All tests fail to reach significance ($p < 0.05$). Adjusted p-values represent the results of applying FDR-BH to address multiple comparisons.

4.3 How does identity inform the mitigations suggested to manage these risks?

Following the risk brainstorming activity, each participant was then asked to identify a possible strategy for managing each of the generated risks (Q13). We performed inductive coding to group these strategies into six categories: Off-Device, Security Hygiene, Don’t Use Product, Don’t Share Data Type, Don’t Share with Party, and Disable/Adjust Other Features. A full codebook with examples and definitions is provided in Appendix B.

4.3.1 Participants Enumerated Multiple Strategies. The most frequently suggested mitigation was, fittingly, Don’t Share with Party, which was entered at least once by 78% of participants (251 participants; 632 responses). This mitigation was across a variety of social contexts, including strangers, advertisers, family, “stalker” (P191), “my partner” (P210), and others. In spite of its popularity, it is interesting to note that “not sharing” may not be an applicable mitigation in the case of vendors and advertisers. With 56% of participants (180 participants; 361 responses), the second most popular mitigation encompasses Off-Device strategies for mitigating risks, including peer support or safety protection mechanisms outside of the tracker such as exercising in groups, exercising in public places, and carrying self defense items. This proportion indicates that non-technical strategies for risk management exceed the expectations of prior work (e.g., observed in 20% of responses in [45]). Don’t Share Data Type, Security Hygiene and Don’t Use Product were suggested by 38%, 27% and 11% of participants, respectively. The codebook in Appendix B includes example quotes and response counts for each. 47% of participants wrote in one or more mitigation to Disable/Adjust Other Features (153 participants; 271 responses). Participants often wanted to use these features to have more granular control over what data was shared and/or with whom their data was shared (e.g., “I check the settings on my devices to make sure location-based ads are disabled. I also try to disable profile-building for advertisers” (P71)). Though their responses rarely named a specific feature that exists in real life, some did convey familiarity with existing features. For example, P63 noted “you can always try to opt-out of personalized ads so you

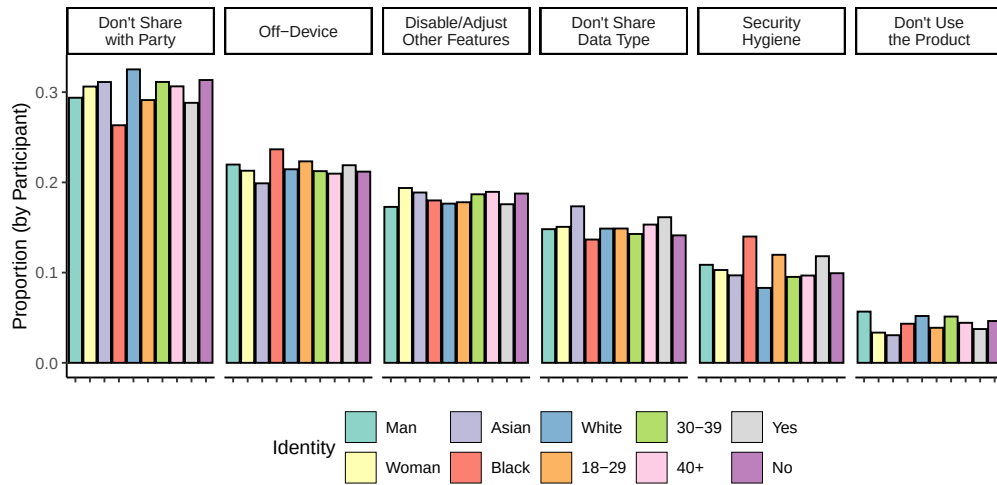


Figure 7: Differences in mitigations by demographics (Q13)—Mitigations compared across different identity groups. Identity Factors are denoted with bar patterns while Identity Levels are color-coded. To simplify comparison between factors with 2 levels (Gender, LGBTQ) and 3 levels (Race, Age), we report the proportion of responses for a given Identity Level that identify a given risk (n=1610). The analysis shows consistent patterns across different demographics, revealing no significant differences in identity.

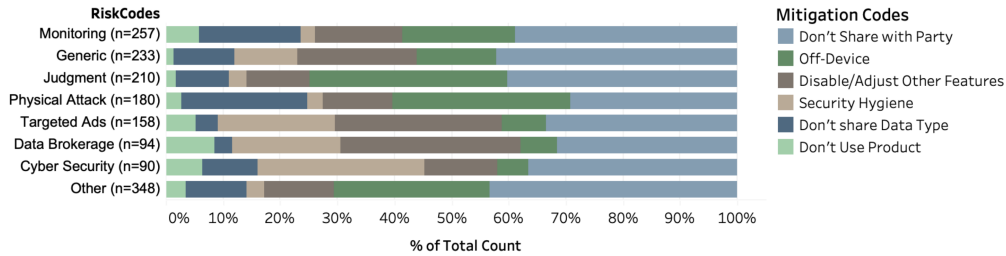


Figure 8: Mitigation distributions across each risk (Q13)—This figure plots an overview of various mitigations suggested by participants to address each risk.

won't be able to see ads that target people like you" and P80 stated "turning off or obfuscating location data."

4.3.2 Mitigation Strategies Considering Risks and Identity. We provide a breakdown of mitigations by identity characteristic in Figure 7. Based on visual inspection, it appears that no significant variation exists in the mitigations employed across different identity groups. To confirm, we repeat the same χ^2 tests from the prior section using the proportion of brainstormed mitigations (by participant). Results are shown in Table 6; as expected, no significant differences ($p < 0.05$) between groups are found.

Figure 8 reports the mitigation distributions across each risk. Many participants, regardless of identity, suggested similar mitigation strategies. Don't Share with Party is common for all risks but the proportions are different. Security Hygiene is in high proportion for Cybersecurity risks. For example, P274, a Black person, suggested a mitigation for "unauthorized access to sensitive health information" (Cybersecurity) is to "regularly update app permissions, use strong authentication method" (Security Hygiene). P278, a white person, suggested a mitigation for "information could

be hacked into and taken by unsavory people" (Cybersecurity) is to "keep your information password protected and continually update it to ensure safety" (Security Hygiene). Such comments reflect a common understanding of cybersecurity practices that applies across different identities.

Off-Device is in high proportion for Judgment. For example, when faced with privacy risks related to Judgment, P275 suggested to mitigate the misuse of health information for judgment or discrimination, a possible solution would be "selective in sharing fitness data on dating apps, consider platforms with robust privacy controls, and prioritize connections with individuals who appreciate you beyond superficial health metrics to foster genuine relationships" (Off-Device). In response to concerns about being criticized by family for not meeting expectations (Judgment), P6 suggested that a possible mitigation would be to "encourage family and friends to keep their opinions to themselves if they come to you" (Off-Device). When faced with privacy risks related to Physical Attack, participants consistently recommended a variety of Off-Device mitigations such as diversifying running routes,

carrying self-defense devices, etc. For example, P264, a man, stated a mitigation for “stalkers or people who would do me harm” (Physical Attack) is to “be vigilant, run a variety of routes, change the area up, and communicate with friends or family about when you start and when you finish.” (Off-Device).

Even across different types of risks, such as Targeted Ads or Data Brokerage, the suggestions remained consistent. A large portion of participants suggested Disable/Adjust Other Features. For instance, P263, an Asian person, suggested to mitigate “the sell my information to other vendors” (Data Brokerage) by “opting out of marketing and advertising data share when available” (Disable/Adjust Other Features). P318, a Black person, advised to mitigate “predatory advertising” (Targeted Ads) is to “be more mindful of and change privacy settings” (Disable/Adjust Other Features). Additionally, when considering risks related to social media data sharing (i.e., Monitoring), some portion of participants recommended choosing not to share certain data types (Don’t Share Data Type). For example, P191, an Asian woman stated: to mitigate a risk of “knowing where you are” (Monitoring) is to “not post on social media regarding my running habits” (Don’t Share Data Type). This uniformity across multiple risk scenarios reinforces the notion that participants were drawing from a shared body of knowledge, likely influenced by widely available risk management advice.

5 Discussion

Although issues of social identity are complex and multifaceted, our findings highlight the influence of participants’ identities on their privacy perceptions when using fitness trackers. We provide evidence that characteristics of participants’ identities inform their comfort sharing fitness data with different parties, and that many participants self-report that their identities inform that comfort. This variance in comfort may be informed, in part, by the perceived risks posed by sharing fitness data; we also find that identity characteristics such as race and age significantly influence those perceptions. However, we do not find evidence that identity plays a prominent role in the strategies they employ to manage risks. We now consider the broader implications of our findings.

5.1 Self-Awareness and Reflection on the Impact of Identity

When exploring comfort levels sharing fitness data, we deliberately asked participants’ to rate their comfort (Q2) before priming them to the topic of identity. However, the follow-up questions “Do you think your <identity characteristic> informs your comfort sharing with any of the above groups of people?” (Q3 – Q9) all explicitly prompt participants to reflect on identity’s role in their prior responses. We find that many patterns in our unprimed former questions are aligned with participants’ qualitative reflections in the primed latter questions. For example, our regression analysis finds that women participants were less comfortable sharing fitness data with interpersonal groups, and were also reported more than men that gender informed their comfort with all/some groups. Associated qualitative responses were similarly well-aligned with sociopolitical issues, for example, “I’m not as afraid as a woman might be about sharing my exact location. Most stalkers and strange

folks seem to be men online targeting women” (P289). Participants’ reflections on the role of identity also provide context for some negative results in the regression analysis. For example, we did not find evidence that participants identifying as LGBTQ* were systematically less comfortable sharing with any group; these participants were also less likely to report that their LGBTQ* status informed their responses than participants that did not identify as LGBTQ*.

Several factors may explain these inconsistencies. One possibility is that our survey instrument lacked the sensitivity to capture nuanced intersections of identity and risk perception, particularly given the abstract nature of the social contexts we presented to participants. For example, participants evaluated their comfort levels through a brainstorming exercise, which may not reflect real-world scenarios. Alternatively, participants may not perceive fitness trackers as exacerbating their lived risks. Previous studies suggest that concerns such as jogging alone at night may not be viewed as linked to fitness tracker usage [12, 16, 28], even though GPS-enabled wearables can expose users to location tracking risks, potentially revealing sensitive information about their routines and making them vulnerable to threats such as stalking or harassment [8, 29, 63].

Despite these limitations, the observed identity-based trends underscore the need for inclusive solutions to fitness tracking privacy. Addressing the specific risks faced by marginalized groups is critical. For example, one participant explicitly referenced menstrual tracking in discussing fitness privacy, highlighting the potential misuse of such data in light of the US Supreme Court’s decision to overturn national abortion protections. This underscores the heightened privacy vulnerabilities faced by individuals whose health-related data could be weaponized against them, aligning with previous reports of menstrual data misuse by state agencies [58]. Given these risks, vendors must prioritize protections for vulnerable groups and consider the specific lived experiences of diverse users when designing privacy-preserving systems.

It is also important to recognize that our sample, which included highly athletic participants, may not fully represent the broader population of fitness tracker users. Athletic participants may exhibit different comfort levels or risk perceptions compared to less active users, particularly regarding data-sharing contexts. Future research should aim to diversify participant demographics to capture a more comprehensive understanding of fitness data privacy concerns across various user groups.

5.2 Contextualizing Our Results in the Sociopolitical Landscape

Many of our findings appear to align with the prevailing sociopolitical challenges faced by different groups, reflecting broader concerns about privacy and data security that are influenced by these challenges. For instance, we observe that women are frequently less comfortable sharing fitness data outside of interpersonal social contexts, and participants from minoritized or marginalized groups were more likely to report that their identity influenced their comfort with sharing fitness data (Section 4.1). Similarly, younger participants and people of color highlighted distinct risks in their responses, suggesting that identity significantly informs risk perception (Section 4.2). On the other hand, certain findings appear to contradict sociopolitical trends reported in prior literature. For

example, while LGBTQ* individuals face documented risks of violent crime and medical service denial [19], we found no evidence that they feel less comfortable sharing fitness data (Section 4.1). Similarly, despite research showing that women are at greater risk of stalking or physically attack compared to men [69], our results did not indicate differences in risk perception or comfort within interpersonal sharing contexts (Section 4.2).

Previous research by Goldfarb and Tucker [23] highlights how privacy preferences evolve over time, with older individuals demonstrating a greater reluctance to disclose personal information compared to younger cohorts. However, the study does not explore how these age-related differences manifest in specific contexts such as fitness tracking, where privacy risks could disproportionately impact older individuals due to health vulnerabilities. While our findings suggest that participants acknowledge such age-specific physical risks, integrating these insights into broader discussions of sociotechnical systems could offer a more comprehensive understanding of privacy preferences.

Similarly, Friedman et al. [20] explored privacy concerns in public spaces and found that women are more likely to voice privacy concerns than men, regardless of whether they are observers or participants. This pattern of heightened privacy awareness among women aligns with research by Hoy and Milne [33], which found that women exhibit stronger privacy preferences on social platforms. However, these studies do not examine how gendered privacy concerns intersect with specific technologies like fitness trackers, where risks such as location sharing may be tied to personal safety. Our research addresses this gap by exploring how gender influences perceptions of risk and privacy in fitness tracking, contributing a gender-sensitive lens to the discourse on privacy design.

In a cross-cultural context, Larson and Medora [38] found that cultural values significantly shape privacy preferences, with American participants expressing a higher need for privacy than their Indian counterparts. Similarly, Bellman et al. [5] identified significant variations in privacy concerns across different countries. However, these studies do not delve into the intersections of culture and individual identity in shaping privacy attitudes. In contrast, Hong et al. [32] showed that African American participants emphasized greater control over personal information, reflecting historical and systemic trust issues. Building on these findings, our study examines the nuanced interplay of race, cultural background, and privacy in fitness tracking ecosystems, addressing a critical gap in identity-informed privacy research.

5.3 Inclusivity and Equity of Fitness Tracking

Our findings suggest that participants' identities do not significantly inform the mitigations they employ to manage risks in sharing fitness data (Section 4.3). A potential explanation for this result is the clustering of mitigation strategies across all parties and risk types, which may dilute identity-driven effects. Given the complexity of our dataset, we prioritized reducing error rates over uncovering (potentially spurious) correlations, leaving open the possibility that nuanced identity-based differences remain undetected. However, the lack of differentiation in mitigations may also reflect a systemic issue: current privacy management strategies are constrained by limited expressivity, failing to accommodate the diversity of

user concerns and risks posed by fitness tracking. As prior work highlights, privacy controls often force binary decisions—accept or reject—rather than offering granular risk management options [22].

This points to the need for privacy and safety technologies that move beyond a one-size-fits-all approach. For example, systems could adopt customizable privacy frameworks that address diverse needs across different identity groups. By incorporating user-centered design principles, vendors can enhance transparency and functionality to support more tailored mitigations. Extending findings from previous studies [45], we observed that 56% of participants consider off-device strategies for managing risks, particularly in response to severe threats like physical attack and monitoring. This trend underscores a desire for broader privacy frameworks that transcend on-device controls, addressing concerns such as third-party data sharing and surveillance.

Future research should explore the design and adoption of off-device strategies, engaging diverse stakeholders to ensure inclusivity and relevance across demographic and sociopolitical contexts. Additionally, studies should utilize how participants' values and critically-oriented design approaches to foreground social values and help define privacy problem spaces, as highlighted in prior work [73]. Such frameworks encourage a more holistic view of privacy that accounts for not only technical aspects but also the social and ethical implications of data usage in technologies like fitness trackers and wearables. These insights can inform educational interventions and systems that correct misunderstandings, promoting safer data-sharing practices.

6 Conclusion

We conduct (to our knowledge) the first systematic study of how key demographic factors that form social identities – Gender, Race, Age, and LGBTQ* status – affect how physically active participants perceive and manage the privacy risks of fitness tracking usage. We discover patterns in (dis)comfort that appear to mirror sociopolitical issues for a number of different groups. Overall, our findings underscore the importance of developing tailored privacy controls that address identity-based differences in risk perception and mitigation strategies. They also emphasize the need for further research to deepen our understanding of how various identity characteristics influence privacy concerns and to improve the effectiveness of privacy management practices. While more work is required, this study demonstrates the need to examine fitness privacy through the lens of identity.

Acknowledgments

This paper results from the SPLICE research program, supported by a collaborative award from the National Science Foundation (NSF) SaTC Frontiers program under award 1955228. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies, either expressed or implied, of NSF. Any mention of specific companies or products does not imply any endorsement by the authors, by their employers, or by the NSF. We thank our anonymous reviewers for their helpful comments in revising this paper. We would like to thank the members of STS Lab for the valuable discussions and feedback.

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A Survey Materials

The following is the full survey instrument, including the Group Comfort Questions, Identity Comfort Questions, Risk Brainstorming Questions, questions to write potential strategies for managing risks, and demographic questions. We present survey in the order that we showed the participants in the online survey. We use the blue italic text to describe the survey flow which was not shown to participants.

Definition of Fitness Trackers

Fitness trackers are body-worn technologies or mobile apps that automatically record physical activity to help users better understand of themselves and their environments. For instance, fitness trackers allow users to track their activities, such as runs or bike rides. Examples of popular fitness trackers include Apple Watch, Samsung Gear, Fitbit, and Garmin. There are also various companion mobile fitness tracking apps such as Strava, Garmin Connect, Apple Health, MapMyRun, and NikeRun. When a user is exercising, these fitness tracking products keep track of statistics such as their blood pressure, pace, speed, elevation, and a map of their exercise route. When the user finishes, they can share their activity with friends or other athletes using the fitness app. If desired, they can share the exercise to social networks like Facebook.

For example, Sam uses the Strava fitness app and Apple Watch to track their exercise routines and heart rate. When they want to go for a run outdoors, they browse public activities posted by other Strava users to discover routes before recording their run and setting off from home. Upon returning home, Sam checks their performance on Strava app. Since they linked their Facebook account to Strava, the run can also be viewed by their friends on Facebook.

- Q1 Once returning from their exercise, what did Sam do? Please re-read the study instruction above if you are not sure.
- ☐ Went to a group workout studio.
 - ☐ Checked their performance on the Strava App.
 - ☐ Had a glass of water.

Perception on Fitness Privacy

The questions on this page ask about your perception on fitness trackers. Think about Sam's experience using fitness trackers as described from the previous page:

Sam uses the Strava fitness app and Apple Watch to track their exercise routines and heart rate. When they want to go for a run outdoors, they browse public activities posted by other Strava users to discover routes before recording their run and setting off from home. Upon returning home, Sam checks their performance on Strava app. Since they linked their Facebook account to Strava, the run can also be viewed by their friends on Facebook.

- Q2 Imagine that your exercise routines were like Sam's, how comfortable would you be sharing data from your fitness tracker with the following groups of people? [(a) Friends, (b) Family, (c) Romantic Partner, (d) Employers, (e) Insurance Company, (f) Advertisers, (g) Fitness Trackers Vendors, (h) General Public]
- ☐ Extremely uncomfortable (1)
 - ☐ Slightly uncomfortable (2)
 - ☐ Neither comfortable nor uncomfortable (3)
 - ☐ Slightly comfortable (4)
 - ☐ Extremely comfortable (5)
- Q3 Do you think your gender informs your comfort in sharing data with any of the above groups of people?
- ☐ All of these groups of people
 - ☐ Some of these groups of people
 - ☐ None of these groups of people
 - ☐ I'm not sure
- Q4 Please explain your answer to the previous question. For example, "HOW" or "WHY" does(n't) your gender inform your comfort? In what ways do your experiences differ from someone of another gender?
-
- Q5 Do you think your race or ethnicity informs your comfort in sharing data with any of the above groups of people?
- ☐ All of these groups of people
 - ☐ Some of these groups of people
 - ☐ None of these groups of people
 - ☐ I'm not sure
- Q6 Please explain your answer to the previous question. For example, "HOW" or "WHY" does(n't) your race or ethnicity inform your comfort? In what ways do your experiences differ from someone of another race or ethnicity?
-
- Q7 Do you think your age informs your comfort in sharing data with any of the above groups of people?
- ☐ All of these groups of people
 - ☐ Some of these groups of people
 - ☐ None of these groups of people
 - ☐ I'm not sure
- Q8 Please explain your answer to the previous question. For example, "HOW" or "WHY" does(n't) your age inform your comfort? In what ways do your experiences differ from someone of another age?
-

- Q9 Does whether or not you are LGBTQ* inform your comfort in sharing data with any of the above groups of people?
- ☐ All of these groups of people
 - ☐ Some of these groups of people
 - ☐ None of these groups of people
 - ☐ I'm not sure

- Q10 Please explain your answer to the previous question. For example, "HOW" or "WHY" does(n't) being LGBTQ* inform your comfort? In what ways do your experiences differ from someone that is(n't) LGBTQ*?
-

Brainstorm Risks

Participants answered the brainstorm risks and mitigations risks section five times.

The following questions ask you to imagine five risks that could arise from sharing data from fitness trackers with different groups of people. Risks are highly individual and can take many forms, including risks to privacy, physical safety, emotional well-being, and security, and can be due to many factors. The "Next" button will be enabled once you fill in all required fields.

- Q11 *This question is repeated to generate 5 potential risks of sharing data from fitness tracking.* Fill in the blanks in the following paragraph to describe a risky situation. When sharing data from fitness trackers with [(a) *Select Group*: Friends, Family, Romantic Partner, Employers, Insurance Company, Advertisers, Fitness trackers vendors, General public] a risk that might arise is [(b) *Enter the risk*]. This is to happen to someone [(c) *Select Likelihood*: Extremely unlikely (1), Somewhat unlikely (2), Neither likely nor unlikely (3), Somewhat likely (4), Extremely likely (5)]. If it did happen, it would be [(d) *Select harmfulness*: Not at all harmful (1), Slightly harmful (2), Somewhat harmful (3), Moderately harmful (4), Extremely harmful (5)].

Mitigating Risks

The following section will ask you to describe strategies to reduce the risks. A variety of strategies could be employed to mitigate risks and concerns.

- Q12 Has this ever happened to you?
- ☐ Yes
 - ☐ Not sure
 - ☐ No
- Q13 What strategies could you use to manage this risk?
-

Demographic Questions

The questions on this page ask about your demographic information.

- Q14 How old are you? Write the number only.
-
- Q15 What is your gender?
- ☐ Woman
 - ☐ Man
 - ☐ Non-binary
 - ☐ Prefer to self-describe:
 - ☐ Prefer not to answer
- Q16 Do you identify as LGBTQ*?
- ☐ Yes
 - ☐ No
 - ☐ Maybe
 - ☐ Prefer not to answer
- Q17 How do you describe your race or ethnic identity? You may select more than one option.
- ☐ Native Hawaiian or Pacific Islander
 - ☐ Asian
 - ☐ American Indian or Alaska Native
 - ☐ Black or African American
 - ☐ White
 - ☐ Not listed above (you may describe if you wish)
 - ☐ Prefer not to respond
- Q18 How would you describe your cultural background? Culture is the shared characteristics of a group of people, which encompasses , place of birth, religion, language, cuisine, social behaviors, art, literature, and music.
- Q19 What is the highest degree you have earned?
- ☐ Less than high school diploma
 - ☐ High school diploma/GED
 - ☐ Vocational qualification
 - ☐ Some college but no degree
 - ☐ Associate degree in college (2-year)
 - ☐ Bachelor's degree in college (4-year)
 - ☐ Master's degree
 - ☐ Doctoral degree
 - ☐ Professional degree (JD, MD)
 - ☐ Prefer not to respond
- Q20 What is your current employment status? (You may select more than one option)
- ☐ Employed full-time
 - ☐ Employed part-time
 - ☐ Self-employed
 - ☐ Student
 - ☐ Retired
 - ☐ Homemaker
 - ☐ Unemployed
 - ☐ Other
 - ☐ Prefer not to respond
- Q21 What is your current annual income?

- | | |
|---|---|
| ☐ US\$ 19,999 or below | ☐ US\$ 60,000 - US\$ 69,999 |
| ☐ US\$ 20,000 - US\$ 29,999 | ☐ US\$ 70,000 - US\$ 79,999 |
| ☐ US\$ 20,000 - US\$ 29,999 | ☐ US\$ 80,000 - US\$ 89,999 |
| ☐ US\$ 30,000 - US\$ 39,999 | ☐ US\$ 90,000 or above |
| ☐ US\$ 40,000 - US\$ 49,999 | ☐ Prefer not to respond |
| ☐ US\$ 50,000 - US\$ 59,999 | |

Q22 What is your current marital status?

- | | |
|--------------------------------|---|
| ☐ Single | ☐ Bereaved |
| ☐ Married | ☐ Other |
| ☐ Divorced | ☐ Prefer not to respond |

Q23 Which of these best describes the general area where you live?

- | | |
|--------------------------------|---|
| ☐ Urban | ☐ Prefer not to respond |
| ☐ Suburban | ☐ I do not know |
| ☐ Rural | |

Q24 How would you describe your level of comfort/experience with computers? level of comfort & level of experience

- ☐ Very poor (1)
- ☐ Below average (2)
- ☐ Average (3)
- ☐ Above average (4)
- ☐ Excellent (5)

Q25 **Feedback**

When you click 'next' below, you will be given a registration code to register your submission back to Prolific and your response will be recorded and included in the research. If you have any further feedback, please let us know.

B Codebooks

Table 7 presents the codebook used for the risks. It contains the risk codes, descriptions, participant examples, response counts, the number of participants who reported each risk at least once, and Cohen’s kappa values. Similar codebook on risks and mitigations can be found in Table 8 and 9 respectively.

Category Name	Option Name	Option Definition	Option Example(s)	Cohen’s Kappa
IDENTITY DOESN’T MATTER	yes - general no or n/a	Explicitly states that the measured identity characteristic (Gender, Race, Age, LGBTQ status) does not matter, or they cannot see why it would matter (to anyone).	“I don’t think it really matters” “race has no bearing on anything in society,” “age is also irrelevant.”	0.76 moderate, 16 disagreements
COMFORT. Multiple codes (MORE+LESS) permitted to indicate “nuanced answer.”	MORE_COMFORT	Response indicates participants report their identity made them more comfortable (or likely) to share fitness tracking data.	“I am white, so for similar reasons as above as to why I would feel more comfortable in sharing my physical running location” (P255)	0.71 moderate, 21 disagreements
	LESS_COMFORT	Response indicates participants report their identity made them less comfortable (or likely) to share fitness tracking data.	“The younger generation growing up with computers and social media seem much more comfortable with sharing their data.” (P255)	
	NONE_COMFORT	No indication of their comfort directionality.	“Who I am attracted to has nothing to do with my concern for privacy” (P254)	
RISK. Multiple codes (MORE+LESS) permitted to indicate “nuanced answer.”	MORE_RISK	Indicates that they are more at risk (sharing or generally)	“As a female, I would be more vulnerable to attack if someone I didn’t know was tracking me.” (P14)	0.84 almost perfect, 11 disagreements
	LESS_RISK	Indicates that they are less at risk (sharing or generally)	“I’m not sure how it relates to my comfort but that could be my privilege as a man.” (P156)	
	NONE_RISK	No indication of their risk directionality.	“Who I am attracted to has nothing to do with my concern for privacy.” (P254)	
RISK-COMFORT. One code.	yes no or n/a	Response explicitly connects comfort to risk with a cause-and-effect statement.	“Being heterosexual, the fact that I am not LGBTQ informs my comfort. I would not be a target of anyone that has biases against such persons.” (P75)	0.77 moderate, 13 disagreements
RISK_SPECIFIC. One code.	yes no or n/a	Describes details of a specific risk.	“immediate dangers to personal safety... there are some states where that information could be used to prosecute me if I or someone I cared about needed healthcare and I didn’t remove my tracker.” (P71)	0.88 almost perfect, 10 disagreements
FACTORS. One code.	yes no or n/a	Participant mentions any unmeasured facet of identity, personality, or demographic. In other words, *not* Gender/Race/Age/LGBTQ.	athletic performance (P97), being comfortable with yourself (P122), generational differences (P175), personal decision (P269)	0.7 moderate, 27 disagreements
PASSING. One code.	yes no or n/a	Participant mentions that (in the real world) it may not be clear they belong to a certain identity group. In other words, they can “pass” as another identity group.	“I’m White passing, so like above, I don’t worry about specific issues, but I am deeply uncomfortable sharing some of this information.” (P320)	1.0 perfect, 0 disagreements
INTERSECTIONALITY. One code.	yes no or n/a	Explicitly mentions (any) overlapping identity (combination of Gender/Race/Age/LGBTQ)	“Lgbtg people do have a risk just as woman do. Hate crimes are always possible.” (P242) “A black trans more says a lot about a person, so would society think, so the less one shares, the better.” (P297)	0.66 moderate, 2 disagreements

Table 7: Codebook of COMFORT

Option Name	Definition	Examples	Counts	Cohen's Kappa
DATA BROKERAGE	Participant's response describes a risk that involves brokers selling data, losing the control of data due to data being sold to another entity	"the data will be used to sell to any company for possible reasons not ok with the people they got the data from." (P49)	94 responses, 80 participants	0.73 moderate (379 disagreements of 1615 responses)
TARGETED ADS	Participant's response describes a risk that involves advertisements, involves not trusting the company's business model.	"personal data is used for advertising purposes (or) receive more targeted advertising" (P081)	158 responses, 134 participants	
CYBERSECURITY	Participant's response describes a risk that involves hackers accessing data or not trusting the company's security practices.	"leaked information" (P084), "data breached" (P214), "information could be hacked into and taken" (P278)	90 responses, 70 participants	
PHYSICAL ATTACK	Participant's response describes a risk that involves criminal activity in the physical world	"armed robbery" (P036), "cases of sexual assault, stalking, or hate crimes" (P285), "judgement and criticism about your fitness routine" (P16), "potential workplace discrimination" (P198), "got jealous" (P236)	180 responses, 138 participants	
JUDGMENT	Participant's response describes a risk that involves social/assumption-based harm.	"strangers can see your route and where you are" (P009), "the employer knowing where you were" (P262), "friends asking what you're doing at certain place" (P296)	210 responses, 146 participants	
MONITORING	Participant's response describes a risk that involves being monitored.	"(vendors) charge different prices" (P001), "Insurance rate increases" (P282)	257 responses, 162 participants	
OTHER	Participant describes other risks that could occur.	"privacy concerns" (P076), "security risks" (P243)	348 responses, 229 participants	
GENERIC	Participant's response describes a generic risk.		233 responses, 117 participants	
NONE (blank)	Participant's response does not describe a risk.	-	-	

Table 8: Codebook of RISKS– Multiple codes per-response permitted.

Option Name	Definition	Examples	Counts	Cohen's Kappa
DON'T SHARE [DATA TYPE]	The mitigation suggested in the response is to not share specific [types of data] or describe specific [types of data] require careful sharing.	"location" (P61), "health information" (P148), "routes you are going to be taking on a run or jog" (P151), "fitness tracker metrics" (P274)	197 responses, 123 participants	0.6 moderate (523 disagreements of 1615 responses)
DON'T SHARE WITH [PARTY]	Participant suggests that one could not share data with certain [party]s.	"strangers", "family", "spouse", "employers", "advertisers", "insurers"	632 responses, 251 participants	
DON'T USE [PRODUCT]	The mitigation suggested in the response is to not use the app / device / tracker (at all).	"I make sure not to use their products" (P011), "I would not use their service from their company" (P049)	62 responses, 37 participants	
DISABLE/ADJUST OTHER FEATURES	The mitigation suggested describes adjusting another feature that may be related to privacy. Includes participant's response conveys actions to "opt out"/ "disable" of sharing that is not reflected in the above codes.	"adjust privacy settings" (P225), "Control who has access to your data by using privacy settings" (P229), "deny permission to the app for tracking this information" (P298)	271 responses, 153 participants	
SECURITY HYGEINE	The mitigation suggested in the response involves using better general technical security practices; may not be related to the fitness tracking domain. Includes mitigations that would make you overall more secure while using the Internet.	"rejecting cookies" (P081), "use VPN to keep identity anonymous" (P220), "install adblockers" (P238), "check terms and conditions as well as privacy terms" (P318), "don't use public wifi" (P321)	133 responses, 87 participants	
OFF-DEVICE	Participant suggests other mitigations that one could perform outside of the tracker.	"exercising in groups, exercising in public places during higher traffic times" (P80), "carrying self defense items like a knife, pepper spray" (P278)	361 responses, 180 participants	
NONE (blank)	Participant's response does not describe a mitigation.	-	-	

Table 9: Codebook of MITIGATIONS– Multiple codes per-response permitted.

C Additional Results

Additional Identity Differences in Risk Likelihood Levels. Figure 9 reports the distribution of Likelihood levels of all risks, with responses distributed across identity characteristics (Gender, Race, Age, and LGBTQ*).

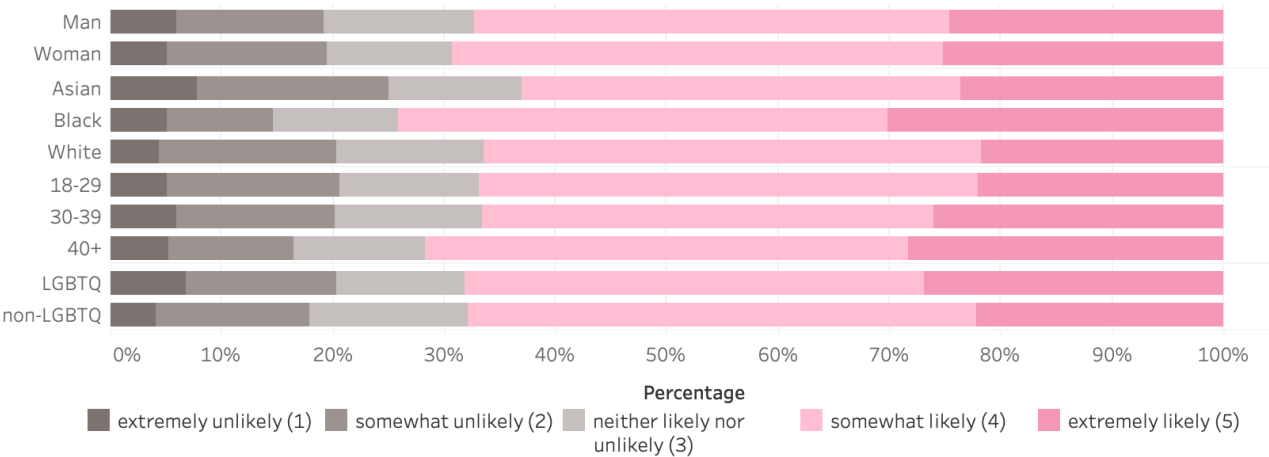


Figure 9: Likelihood level (Q11.(c)) distribution across identity characteristics (gender, race, age, LGBTQ*) for all risks.

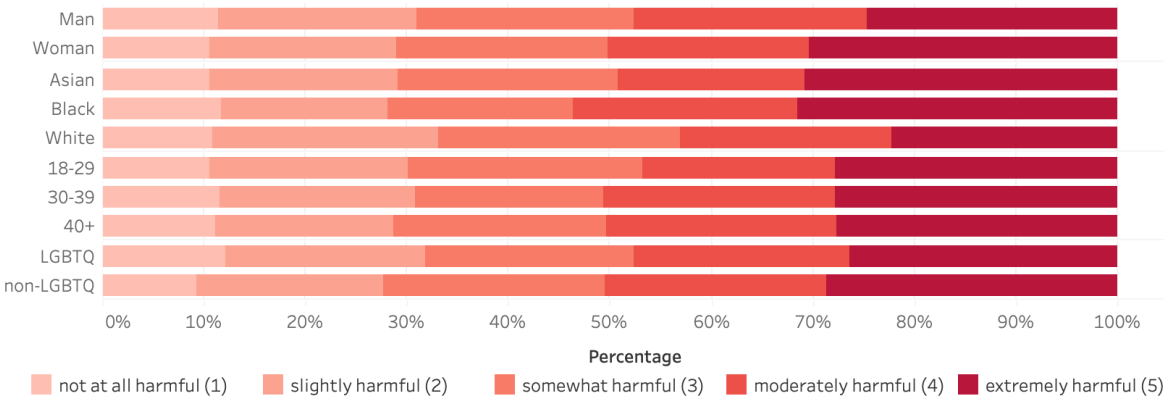


Figure 10: Harmfulness level (Q11.(d)) distribution across identity characteristics (gender, race, age, LGBTQ*) for all risks.

Additional Identity Differences in Risk Harmfulness Levels. Figure 10 reports the distribution of Likelihood levels of all risks, with responses distributed across identity characteristics (Gender, Race, Age, and LGBTQ*).

C.1 Recruitment

Gender	Count	is LGBTQ*?	Count
Man	209	Yes	63
Woman	204	No	339
Non-binary	1	Prefer not to answer	6
Prefer not to say	3	Maybe	9
Age	Count	Race/Ethnicity	Count
18-29	135	Asian	44
30-39	132	Black	33
40+	150	White	310
		Native Hawaiian or Pacific Islander	4
		Prefer not to answer	8
		Not listed above	9

Table 10: Summary of demographic information of the participant sample based on self-reported data from a preliminary data collection ($N=417$).